Universal Structural Selection Benchmarks

Testing MAAT Against Energy, Stability, Random, and Domain-Specific Baselines

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Abstract

Paper 48 derived the exponential structural-selection measure

$$P[X] \propto \exp[-F_{\text{struct}}[X]]$$

from maximum entropy under fixed structural-defect constraints. That theorem establishes the form of the measure, but not the universality of any particular defect basis. The present paper therefore defines a reproducible benchmark protocol for the next step: testing whether MAAT-style structural functionals add predictive information beyond simpler alternatives.

We aggregate existing benchmark outputs across seven domains: nonlinear fields, string-landscape path selection, cosmology, SAT/constraint systems, safety-boundary calibration, Standard-Model-like constant selection, and quantum measurement. Each domain is audited against six competitor classes: energy-only, entropy/activity-only, stability-only, random or shuffled nulls, domain-specific simple baselines, and cross-domain or holdout transfer.

The resulting registry is intentionally mixed. Field and string benchmarks already contain complete energy-only competitor failures. Cosmology has null/permutation protocols but remains diagnostic rather than decisive. SAT currently behaves as a stress test rather than a success, and quantum measurement remains missing. Across the current matrix, the benchmark-readiness index is

$$0.3816,$$

with 8 complete, 13 partial, and 17 missing competitor-test cells. This low value is not presented as a success metric. It is an audit warning: the programme is still early empirically, and the universal claim is not yet supported at the standard required for a mature cross-domain theory.

Thus the present paper does not claim universal proof by evidence. Instead, it converts the universality claim into a concrete, reproducible, and falsifiable benchmark programme: the necessary next step after the formal MaxEnt derivation.

1 Purpose and Status

The maximum-entropy derivation fixes a formal question: if a system is selected under finite information about structural defects, the least-biased measure is exponential in an additive structural functional. In its linear form,

$$F_{\text{struct}}[X] = \sum_a \lambda_a d_a[X], \qquad P[X] = Z^{-1} e^{-F_{\text{struct}}[X]}. \tag{1}$$

In its bounded support form,

$$F_{\text{MAAT}}[X] = -\sum_a \lambda_a \log(\epsilon + \Gamma_a[X]), \qquad \Gamma_a[X] = \frac{1}{1 + d_a[X]}. \tag{2}$$

This does not finish the universality problem. The remaining question is empirical and methodological:

Does the same structural-selection form outperform simpler rankings across independent domains?

This paper is an audit and benchmark-protocol paper. It is not a new solver, not a new cosmological fit, and not a proof that MAAT is fundamental. It identifies where the existing evidence is strong, partial, or missing.

2 Benchmark Logic

A universal structural-selection principle must survive comparison against simple competitors. For each domain, the minimum benchmark set is:

- **energy-only ranking**, where energy or action alone is used;
- **entropy/activity-only ranking**, where dynamical richness alone is used;
- **stability-only ranking**, where robustness alone is used;
- **random or shuffled nulls**, where defect–target associations are broken;
- **domain-specific baselines**, such as LCDM, solver runtime regressions, or standard landscape control quantities;
- **cross-domain or holdout transfer**, where parameters are not tuned to the observable being evaluated.

The benchmark does not require MAAT to win every test. It requires failures to be visible. A negative SAT result, for example, is more useful than an untested success claim because it identifies where the current defect representation is not yet sufficient.

3 Aggregator Method

The Paper 49 aggregator reads existing repository outputs and classifies each domain as:

- `complete`: an explicit competitor or null test exists;
- `partial`: a diagnostic or indirect competitor exists;
- `missing`: the benchmark class is absent;
- `n/a`: the competitor class is not natural for that domain.

The benchmark-readiness index is defined as

$$R_{\text{bench}} = \frac{2N_{\text{complete}} + N_{\text{partial}}}{2(N_{\text{complete}} + N_{\text{partial}} + N_{\text{missing}})}, \tag{3}$$

excluding `n/a` entries. This is not a physical observable. It is an audit score measuring how much of the falsification matrix has been filled, and it should be read as secondary to the individual competitor tests. An energy-only defeat in a controlled field or string benchmark is more important than a formally complete but weak random-null entry. Future versions should therefore replace this unweighted index by a weighted scorecard in which direct competitor failures carry more weight than checklist coverage.

4 Domain Evidence Registry

Table 1: Domain-level evidence registry generated by the Paper 49 aggregator.

Domain	**Competitor**	**Primary result**	**Status**
Nonlinear fields	energy-only	$F_{\mathrm{worst}}/F_{\mathrm{best}} = 7.897$ at only 0.0051% energy spread	complete
String landscape	energy-only path action	top-20 structural/energy edge overlap 0/20; $\rho(\lvert E\rvert, F_{\mathrm{MAAT}}) = 0.1271$	complete
Cosmology	redshift-shuffled null	blind residual correlation $\rho = 0.4286$, permutation $p = 0.1456$	partial
SAT/constraints	base regression	base $R^2 = -22.8763$, v1.2.1 $R^2 = -30.8525$	partial / stress test
AI/safety boundary	non-boundary balance	R share $= 0.3917$, $\lambda_R = 8.0782$	partial
SM-like constants	holdout / random UV	mean holdout $\lvert\log_{10}\text{error}\rvert = 0.3796$, worst factor 5.884	partial
Quantum measurement	decoherence-only	not implemented	missing

5 Strongest Current Evidence

The two strongest results are those where the main simple alternative is directly and visibly defeated. They should be separated from the weaker, partial diagnostics:

Table 2: Current hard competitor defeats.

Domain	**Simple competitor**	**Observed separation**
Fixed-energy nonlinear fields	Energy-only ranking	Near-equal-energy configurations differ by $F_{\mathrm{worst}}/F_{\mathrm{best}} = 7.897$ while their energy spread is only 0.0051%.
Reduced KKLT path graph	Energy-only path action	Top-20 structural and energy-selected transition edges have 0/20 overlap; the selected structural path cost is 0.907889 versus 1.719058 for the energy-action path to the same target.

These are not yet a universal proof, but they are the present nontrivial signal: structural ranking is demonstrably not identical to energy ranking in two independent settings. The two complete energy-only defeats are particularly noteworthy. In fixed-energy field configurations, a structural score difference of nearly 8× occurs at sub-permille energy variation. In the reduced KKLT landscape, the top 20 structural transitions share zero overlap with the top 20 energy-action transitions. Together, these results show that structural selection is not merely a proxy for energy or ordinary stability.

6 Readiness Matrix

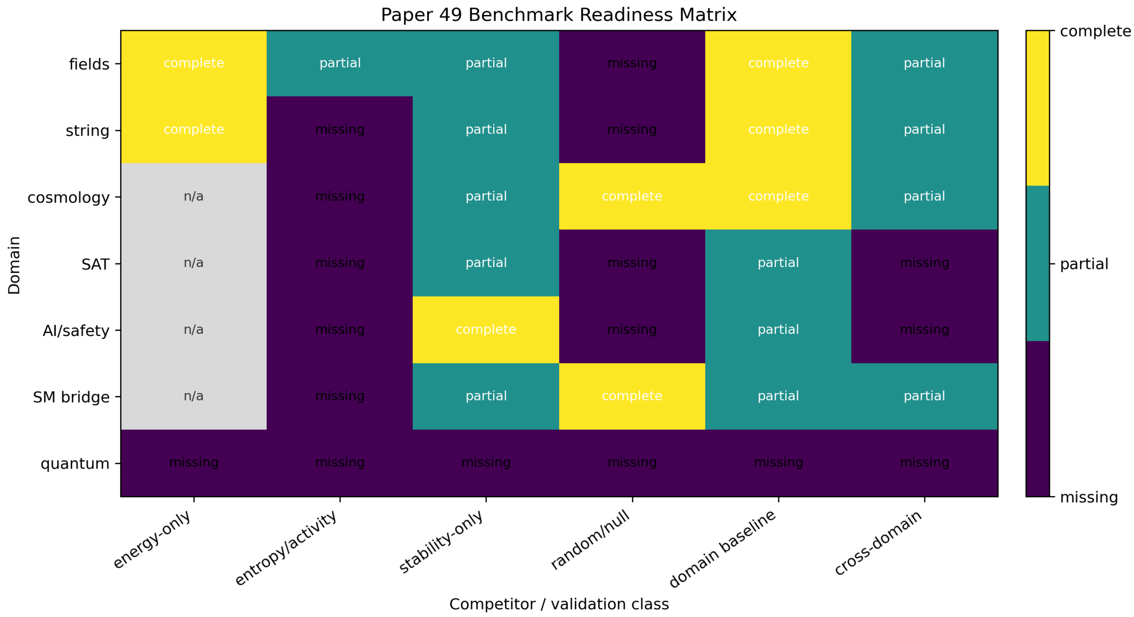

Figure 1: Benchmark-readiness matrix. The strongest completed competitor tests are currently energy-only failures in the field and string-landscape domains. Entropy/activity-only, stability-only, and random/null competitors are not yet standardized across all domains.

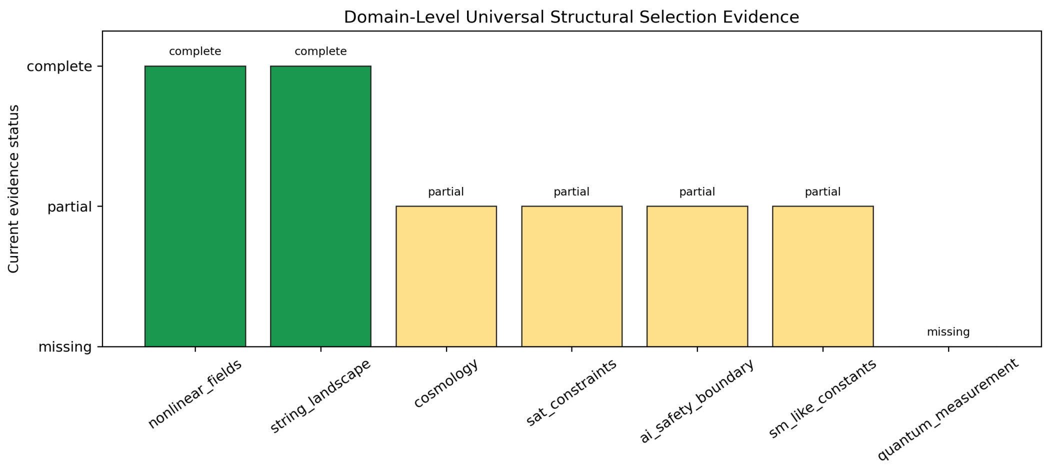

Figure 2: Domain-level evidence status. Two domains currently have complete evidence against their main simple competitor, four are partial, and one required domain is missing.

The generated summary is:

Quantity	Value
Complete domain evidence	2
Partial domain evidence	4
Missing domain evidence	1
Complete matrix cells	8
Partial matrix cells	13
Missing matrix cells	17
Benchmark-readiness index	0.3816

7 Interpretation

The current evidence supports a cautious conclusion:

> *MAAT-style structural selection is already nontrivial in domains where energy-only ranking can be tested directly, but universal status is not yet earned.*

The strongest present results are:

- fixed-energy field configurations can have nearly identical total energy while receiving very different structural scores;
- reduced KKLT transition graphs produce structural paths that differ sharply from energy-only path ordering;
- cosmology has reproducible blind and null-test protocols, but the signals remain diagnostic;
- SAT and quantum measurement are the two most important open stress tests.

The SAT result is deliberately included even though it is not a success. Its poor cross-validated performance indicates that the current SAT feature map does not yet close the universality claim. This is exactly the kind of failure mode a serious benchmark programme must retain.

8 SAT as an Achilles Test

SAT and related constraint systems are especially important because they are objective, non-physical, and far from the field-theory examples that motivated much of the framework. The current SAT result is therefore not a minor weakness. It is the clearest warning that the present defect basis is not yet domain-independent.

There are several plausible failure modes:

- the available SAT features may not capture clause-variable geometry, backbone structure, or solver-specific branching hardness;
- the current H, B, S, V proxies may be too coarse for combinatorial phase transitions;
- runtime is a noisy target because it depends on solver heuristics, not only on instance structure;
- the v1.2.1 closure may require graph-local or scale-dependent defects rather than global summary features.

The next SAT benchmark should therefore not merely refit the existing columns. It should construct explicit graph and solver-independent diagnostics, such as constraint-graph expansion, implication-core depth, backbone fraction, community modularity, clause-variable interaction cycles, backdoor density, and covariance conditioning across local neighborhoods. If those additions still fail against simple solver baselines, the universality claim should be weakened accordingly.

9 Cross-Domain Transfer

The most important future test is not another single-domain fit. It is cross-domain transfer:

> *Can a defect architecture calibrated in one domain retain predictive value in a structurally different domain?*

The existing SM-like holdout benchmark is an early version of this idea: direct terms are removed and the remaining structural terms are tested for basin-level prediction. However, the current result is still partial. The mean holdout error remains finite but not precision-level, and the worst factor is approximately 5.884.

For a stronger universality claim, future transfer tests should include:

- field-to-cosmology transfer of defect weights;
- SAT-to-physics transfer of covariance-geometry diagnostics;
- string-to-field transfer of constraint-balance sectors;
- quantum pointer-state tests using no parameters trained on the measurement benchmark itself.

10 Roadmap for Papers 50–52

The next papers should reduce meta-discussion and increase hard signal. A concrete sequence is:

Next paper	**Immediate target**
Paper 50	**SAT Structural Hardness II:** new graph-local defect sectors and a solver-independent feature map, tested against clause-density, backbone, backdoor-density, and runtime baselines.
Paper 51	**Quantum Pointer-State Selection:** a minimal decoherence model testing whether structural support predicts pointer-basis robustness better than decoherence rate or stability alone.
Paper 52	**Cross-Domain Transfer:** freeze a defect architecture in one domain and evaluate it in another without retuning, using a predeclared score and shuffled-defect nulls.

This roadmap is the operational continuation of the present audit. If Papers 50–52 fail, the universal interpretation should be narrowed. If they succeed, Paper 48's MaxEnt measure gains genuine cross-domain evidential support.

11 Falsification Criteria

The universal structural-selection programme should be considered weakened if:

- F_{MAAT} repeatedly collapses to energy-only ranking;
- entropy/activity alone matches the same target rankings;
- stability-only ranking reproduces all MAAT gains;
- shuffled defects perform as well as structured defects;
- cross-domain transfer fails systematically;

- quantum measurement and SAT remain permanently unclosed.

Conversely, the programme is strengthened if the same exponential support architecture repeatedly adds predictive information after these competitors are included.

12 Reproducibility

Repository:

https://github.com/Chris4081/structural-selection-principle/tree/main/experiments/universal_structural_selection_benchmarks_paper49

Run:

```
python3 universal_structural_selection_benchmarks_paper49.py
```

The script writes:

- `outputs_paper49/paper49_domain_evidence_registry.csv`;
- `outputs_paper49/paper49_benchmark_readiness_matrix.csv`;
- `outputs_paper49/paper49_summary.json`;
- `outputs_paper49/fig1_benchmark_readiness_matrix.png`;
- `outputs_paper49/fig2_domain_evidence_status.png`.

This paper aggregates derived repository outputs. External scientific data used by the underlying experiments remain attributed in their respective papers and experiment folders. No new external dataset is redistributed here.

13 Conclusion

Paper 48 derived the structural-selection measure from maximum entropy. Paper 49 turns the remaining universality claim into an explicit benchmark matrix.

The current result is deliberately not a victory declaration. It is a map: field and string benchmarks already defeat energy-only competitors; cosmology has reproducible but still diagnostic null protocols; AI/safety and SM-like constants remain partial; SAT exposes a real weakness; and quantum measurement is still missing.

This is the right status for a developing universal principle. The measure has a formal derivation. The next standard is not rhetoric, but repeated survival against simple competitors across independent systems. In this sense Paper 49 is not the endpoint of the universality programme. It is the checklist that makes the next failures or successes scientifically legible.

References

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